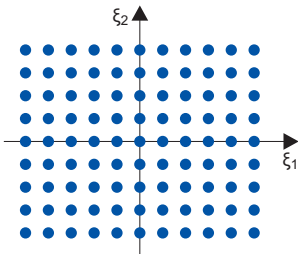


Ohne Erregung und ein reeller EW

$\underline{\lambda} = 0$ und Matrix $\mathbf{A} = \mathbf{0}$

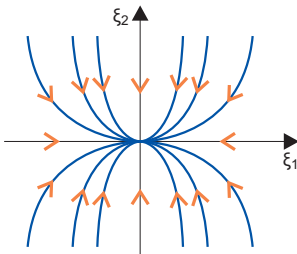
- stabil
- Ebene von Ruhelagen



Ohne Erregung und zwei reelle EW

$\underline{\lambda}_2 < \underline{\lambda}_1 < 0$

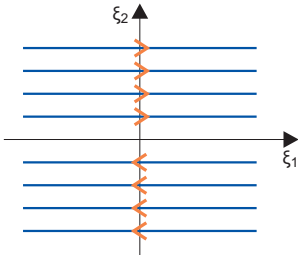
- asymptotisch stabil
- Knoten 2. Art



ZV3

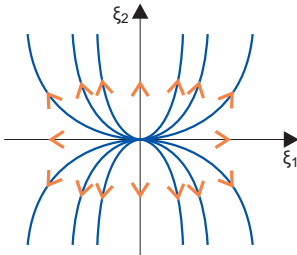
$\underline{\lambda} = 0$ und Matrix $\mathbf{A} \neq \mathbf{0}$

- instabil



$0 < \underline{\lambda}_1 < \underline{\lambda}_2$

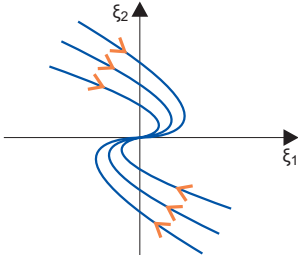
- instabil
- Knoten 2. Art



ZV3

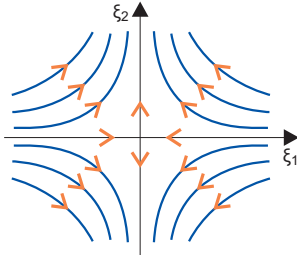
$\underline{\lambda} < 0$ und $\mathbf{A} \neq \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$

- asymptotisch stabil
- Knoten 3. Art



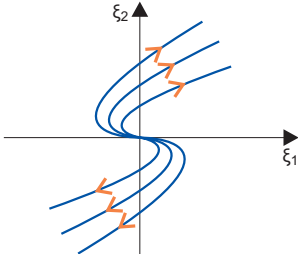
$\underline{\lambda}_1 < 0 < \underline{\lambda}_2$

- instabil
- Sattelpunkt



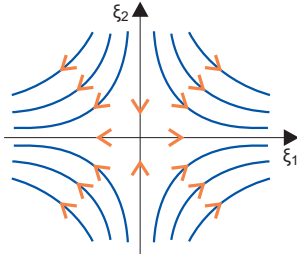
$\underline{\lambda} > 0$ und $\mathbf{A} \neq \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$

- instabil
- Knoten 3. Art



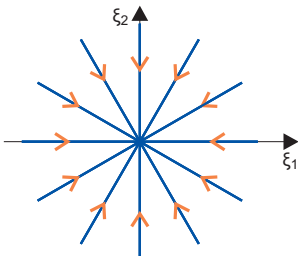
$\underline{\lambda}_2 < 0 < \underline{\lambda}_1$

- instabil
- Sattelpunkt



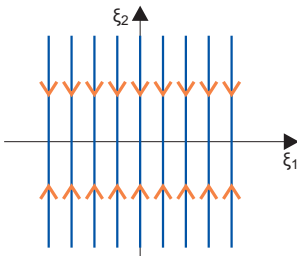
$\underline{\lambda} < 0$ und $\mathbf{A} = \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$

- asymptotisch stabil
- Knoten 1. Art



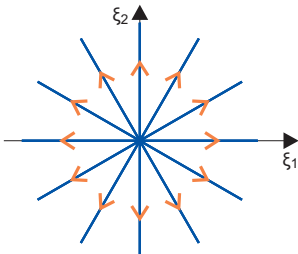
$\underline{\lambda}_2 < 0 = \underline{\lambda}_1$

- ~~- stabil~~
- instabil



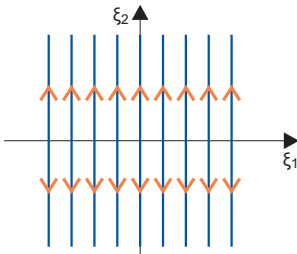
$\underline{\lambda} > 0$ und $\mathbf{A} = \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$

- instabil
- Knoten 1. Art



$\underline{\lambda}_1 = 0 < \underline{\lambda}_2$

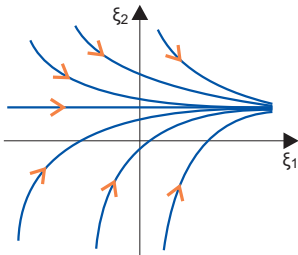
- instabil



Mit Erregung und reellen EW

$\underline{\lambda}_2 < 0 = \underline{\lambda}_1$

- instabil

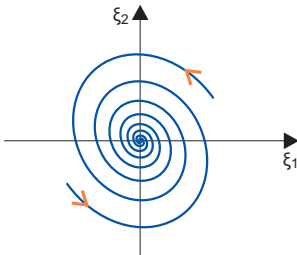


Ohne Erregung und komplexen EW

$\underline{\alpha} < 0; \underline{\beta} > 0$

$\lambda_{1,2} = \alpha \pm j\beta$

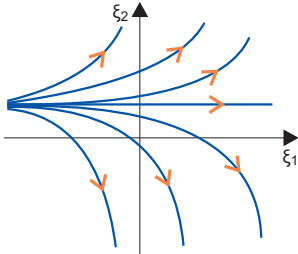
- asymptotisch stabil
- Strudelpunkt
- Drehsinn in x_1/x_2 -Ebene evtl. anders!



ZV2

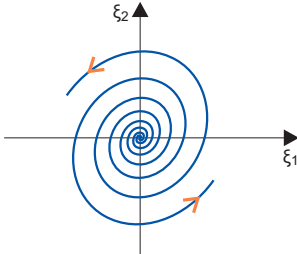
$\underline{\lambda}_1 = 0 < \underline{\lambda}_2$

- instabil



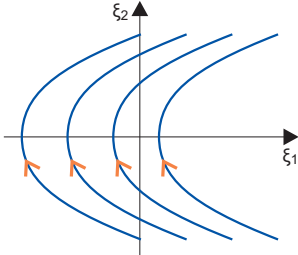
$\underline{\alpha} > 0; \underline{\beta} > 0$

- instabil
- Strudelpunkt
- Drehsinn in x_1/x_2 -Ebene evtl. anders!



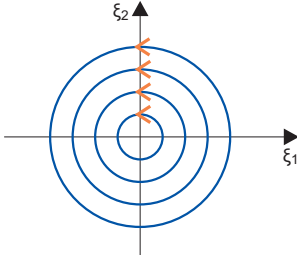
$\underline{\lambda}_1 = \underline{\lambda}_2 = 0$ und $\mathbf{A} \neq \mathbf{0}$

- instabil



$\underline{\alpha} = 0; \underline{\beta} > 0$

- ~~- instabil~~
- ~~- stabil~~
- Wirbelpunkt
- Drehsinn in x_1/x_2 -Ebene evtl. anders!



ZV1